

2.4 - Mid of 3.3.

classical computing	quantum computing.
classical bits	2-dim quantum sys. described by
$\{0, 1\}$	the qubit space $\mathbb{C}^2$
	state $ \psi\rangle \in \mathbb{C}^2$.
	$\rightarrow$ allow states $a 0\rangle + b 1\rangle$ ($ a ^2 + b ^2 = 1$).
information storage small	big
Avoiding state changing interactions easy	hard.

Definition

1. Qubits: quantum mechanical sys. described by 2-dim Hilbert sp. $\mathbb{C}^2$

2. Qubit space: $\mathbb{C}^2$.

$\hookrightarrow$ Select ONB $\{|0\rangle, |1\rangle\}$, which consists of eigenstates of $\hat{\sigma}_z$.

$$\hat{\sigma}_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \Rightarrow \hat{\sigma}_z |0\rangle = |0\rangle \\ \hat{\sigma}_z |1\rangle = -|1\rangle.$$

Corollary 2.29.

A measurement of $\hat{\sigma}_z$ on a qubit yields 1 or -1 as observed val., and projects the qubit in the eigenstate $|0\rangle$ or $|1\rangle$ resp.

→ We represent classical bit as a qubit state.

Table 2.1 Representation of classical bits by qubits

Observed value of σ_z	qubit state	Represented classical bit value
+1	$ 0\rangle = \uparrow_z\rangle$	0
-1	$ 1\rangle = \downarrow_z\rangle$	1

* Parametrization of pure/mixed qubit state.

1. Pure states.

$$|\psi\rangle \in \mathbb{C}^2.$$

$$\Rightarrow |\psi\rangle = a|0\rangle + b|1\rangle, \quad |a|^2 + |b|^2 = 1.$$

$$a = e^{i\alpha} \cos \frac{\theta}{2}, \quad b = e^{i\beta} \sin \frac{\theta}{2}.$$

$$|\psi\rangle = e^{i\alpha} \cos \frac{\theta}{2} |0\rangle + e^{i\beta} \sin \frac{\theta}{2} |1\rangle.$$

$$e^{-i\frac{\alpha+\beta}{2}} |\psi\rangle = e^{i\frac{\alpha-\beta}{2}} \cos \frac{\theta}{2} |0\rangle + e^{-i\frac{\alpha-\beta}{2}} \sin \frac{\theta}{2} |1\rangle$$

$$|\psi(\theta, \phi)\rangle = e^{-i\frac{\phi}{2}} \cos \frac{\theta}{2} |0\rangle + e^{i\frac{\phi}{2}} \sin \frac{\theta}{2} |1\rangle.$$

$$(\because \phi := \beta - \alpha).$$

(Q) What observable has this state as eigenstate?

Notation

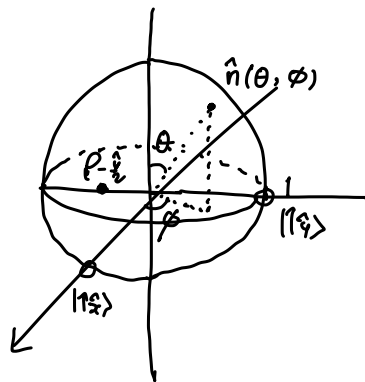
$$\text{For } \alpha = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} \in \mathbb{R}^3,$$

$$\alpha \cdot \sigma := \sum_{i=1}^3 \alpha_i \sigma_i = \begin{pmatrix} a_3 & a_1 - i a_2 \\ a_1 + i a_2 & -a_3 \end{pmatrix}$$

not usual scalar prod.

$$\text{Let } \hat{n} = \hat{n}(\theta, \phi) = \begin{pmatrix} \sin\theta \cos\phi \\ \sin\theta \sin\phi \\ \cos\theta \end{pmatrix}$$

$$\hat{n} \cdot \sigma = \begin{pmatrix} \cos\theta & e^{-i\phi} \sin\theta \\ e^{i\phi} \sin\theta & -\cos\theta \end{pmatrix}$$



: observable 'spin in direction $\hat{n}$ '

$$\hat{n} \cdot \sigma |\psi(\theta, \phi)\rangle = |\psi(\theta, \phi)\rangle.$$

$\therefore \hat{n} \cdot \sigma$: observable that has qubit state $|\psi(\theta, \phi)\rangle$ as an eigenstate w. eigenval. 1.

2. Mixed state.

$$\rho = \begin{pmatrix} a & b \\ c & d \end{pmatrix}; \text{ density operator}$$

$$\Rightarrow 1. \rho^* = \rho \Rightarrow \begin{pmatrix} \bar{a} & \bar{c} \\ \bar{b} & \bar{d} \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \Rightarrow \begin{matrix} a, d \in \mathbb{R} \\ b = \bar{c} \end{matrix}$$

$$2. \text{tr}(\rho) = 1 \Rightarrow a + d = 1$$

for $x_3 \in \mathbb{R}$,

$$3. \rho \geq 0$$

$$\begin{cases} a = \frac{1+x_3}{2} \\ d = \frac{1-x_3}{2} \\ \begin{cases} x_1 := 2\operatorname{Re}(c) \\ x_2 := 2\operatorname{Im}(c) \end{cases} \end{cases}$$

$\Rightarrow$ For $\mathbb{X} = (x_1, x_2, x_3)$,

$$\rho_{\mathbb{X}} = \frac{1}{2} \begin{pmatrix} 1+x_3 & x_1 - ix_2 \\ x_1 + ix_2 & 1-x_3 \end{pmatrix} = \frac{1}{2} (\mathbb{1} + \mathbb{X} \cdot \sigma)$$

$$\downarrow \det(\rho_{\mathbb{X}} - \lambda \mathbb{1}) = \begin{vmatrix} \frac{1+x_3}{2} - \lambda & x_1 - ix_2 \\ x_1 + ix_2 & \frac{1-x_3}{2} - \lambda \end{vmatrix} = 0$$

$$\Leftrightarrow \lambda = \frac{1 \pm |\mathbb{X}|}{2} \geq 0$$

$$1 - |\mathbb{X}| \geq 0.$$

$$\therefore |\mathbb{X}| \leq 1, \quad \text{i.e. } \mathbb{X} \in \mathcal{B}_{\mathbb{R}^3}^1 \quad \text{"Bloch representation"}$$

• All edge pts $|\mathbb{X}|=1$ correspond to pure states.

Pf). Claim) For $a, b \in \mathbb{R}^3$, $(a \cdot \sigma)(b \cdot \sigma) = (a \cdot b) \mathbb{1} + (a \times b) \cdot \sigma$

$$\begin{aligned} \text{Pf of claim). } (a \cdot \sigma)(b \cdot \sigma) &= \sum_{i,j=1}^3 a_i b_j \sigma_i \sigma_j \\ &= \sum_{i,j=1}^3 a_i b_j (\delta_{ij} \mathbb{1} + i \epsilon_{ijk} \sigma_k) \\ &= \sum_{i=1}^3 a_i b_i \mathbb{1} + i \left(a_1 b_2 \epsilon_{123} \sigma_3 + a_2 b_1 \epsilon_{213} \sigma_3 \right. \\ &\quad \left. + a_1 b_3 \epsilon_{132} \sigma_2 + a_3 b_1 \epsilon_{312} \sigma_2 \right. \\ &\quad \left. + a_2 b_3 \epsilon_{231} \sigma_1 + a_3 b_2 \epsilon_{321} \sigma_1 \right) \end{aligned}$$

$$+ a_2 b_3 \cancel{\varepsilon_{231}} \sigma_1 + a_3 b_2 \cancel{\varepsilon_{321}} \sigma_1$$

$$= (a \cdot b) \mathbb{1} + i (a \times b) \cdot \sigma$$

$$\rho_x^2 = \rho_x = \frac{1}{2} (\mathbb{1} + x \cdot \sigma)$$

$$\Leftrightarrow \frac{1}{4} (\mathbb{1} + x \cdot \sigma)^2 = \frac{1}{2} (\mathbb{1} + x \cdot \sigma)$$

$$\Leftrightarrow \mathbb{1} + \cancel{2x \cdot \sigma} + \underbrace{(x \cdot \sigma)(x \cdot \sigma)}_{(x \cdot x) \mathbb{1} + (x \times x) \cdot \sigma} = 2\mathbb{1} + \cancel{2x \cdot \sigma}$$

$$\Leftrightarrow (|x|^2 - 1) \mathbb{1} = 0$$

$$\Leftrightarrow |x| = 1.$$

* Unitary operators on $\mathbb{H}$.

• For $A^2 = \mathbb{1}$, $\alpha \in \mathbb{R}$, define

$$\begin{aligned} e^{i\alpha A} &:= \sum_{n=0}^{\infty} \frac{(i\alpha)^n}{n!} A^n \\ &= \sum_{k=0}^{\infty} \frac{(i\alpha)^{2k}}{(2k)!} A^{2k} + \sum_{j=0}^{\infty} \frac{(i\alpha)^{2j+1}}{(2j+1)!} A^{2j+1} \\ &\quad \underbrace{\sum_{k=0}^{\infty} \frac{(-1)^k \alpha^{2k}}{(2k)!}}_{\cos \alpha} \underbrace{\mathbb{1}}_{\mathbb{1}} + \underbrace{\sum_{j=0}^{\infty} \frac{(-1)^j \alpha^{2j+1}}{(2j+1)!}}_{i \sin \alpha} \underbrace{A}_{A} \end{aligned}$$

$$= \cos \alpha \mathbb{1} + i \sin \alpha A.$$

- Given $\hat{n} \in S^1_{\mathbb{R}^3}$, $\alpha \in \mathbb{R}$, the rotation around $\hat{n}$ by angle α on $\mathcal{H}$ (i.e. spin rotation) is

$$D_{\hat{n}}(\alpha) := e^{-i \frac{\alpha}{2} \hat{n} \cdot \sigma}$$

$$= \cos\left(-\frac{\alpha}{2}\right) \mathbb{1} + i \sin\left(-\frac{\alpha}{2}\right) \hat{n} \cdot \sigma$$

$$\left(\because (\hat{n} \cdot \sigma)(\hat{n} \cdot \sigma) = (\hat{n} \cdot \hat{n}) \mathbb{1} + (\hat{n} \times \hat{n}) \cdot \sigma \right.$$

$$= \mathbb{1} \quad \left. \right)$$

$$= \cos \frac{\alpha}{2} \mathbb{1} - i \sin \frac{\alpha}{2} \hat{n} \cdot \sigma.$$

$$D_{\hat{n}}(\alpha)^* = \cos \frac{\alpha}{2} \mathbb{1} + i \sin \frac{\alpha}{2} \hat{n} \cdot \sigma.$$

$$\left(\because \hat{n} \cdot \sigma = \sum_{i=1}^3 n_i \sigma_i, \quad \sigma_i \text{ was self-adjoint} \right)$$

$$D_{\hat{n}}(\alpha) D_{\hat{n}}(\alpha)^* = \mathbb{1}$$

$$\Rightarrow D_{\hat{n}}(\alpha) \in U(\mathcal{H}).$$

* Some expressions of $U \in U(4)$.

$$1. \exists \alpha, \beta, \gamma, \delta \in \mathbb{R} \text{ st } U = e^{i\alpha} D_{\frac{\gamma}{2}}(\beta) D_{\frac{\delta}{2}}(\gamma) D_{\frac{\alpha}{2}}(\delta).$$

$\nwarrow \nearrow$
 spin rotation around
 y -axis / z -axis

$$2. \exists A, B, C: \text{ operators, } \alpha \in \mathbb{R} \text{ st } \begin{cases} U = e^{i\alpha} A \sigma_x B \sigma_z C \\ ABC = \mathbb{1} \end{cases}$$

Sketch of Pf). By 1, $U = e^{i\alpha} D_{\frac{\gamma}{2}}(\beta) D_{\frac{\delta}{2}}(\gamma) D_{\frac{\alpha}{2}}(\delta).$

$$\text{Set } A := \cancel{D_{\frac{\gamma}{2}}(\beta) D_{\frac{\delta}{2}}(\frac{\gamma}{2})}$$

$$B := \cancel{D_{\frac{\delta}{2}}(-\frac{\gamma}{2}) D_{\frac{\alpha}{2}}(-\frac{\delta+\beta}{2})}$$

$$C := \cancel{D_{\frac{\alpha}{2}}(-\frac{\delta-\beta}{2})} \rightarrow \underline{D_{\frac{\alpha}{2}}(-\beta)}.$$

$$\Rightarrow ABC = \mathbb{1}.$$

$$\Rightarrow e^{i\alpha} A \sigma_x B \sigma_z C = \dots = U$$

$$3. \exists \hat{n} \in S_{\mathbb{R}^3}^1, \alpha, \frac{\xi}{2} \in \mathbb{R} \text{ st. } U = e^{i\alpha} D_{\hat{n}}(\frac{\xi}{2})$$

$$4. U = z_0 \mathbb{1} + \mathbf{z} \cdot \boldsymbol{\sigma} = \sum_{\alpha=0}^3 \frac{z_{\alpha}}{2} \sigma_{\alpha} \text{ for some } z_0 \in \mathbb{C}, \mathbf{z} \in \mathbb{C}^3$$

st $|z_0|^2 + |\mathbf{z}|^2 = 1$

Pf) By 3, $U = e^{i\alpha} D_{\hat{n}}(\frac{\xi}{2}).$

$$= e^{i\alpha} \left(\cos \frac{\xi}{2} \mathbb{1} - i \sin \frac{\xi}{2} \hat{n} \cdot \boldsymbol{\sigma} \right)$$

$$= \underline{e^{i\alpha} \cos \frac{\xi}{2} \mathbb{1}} - \underline{i e^{i\alpha} \sin \frac{\xi}{2} \hat{n} \cdot \boldsymbol{\sigma}}$$

$$\begin{matrix} \parallel \\ \mathbf{z}_0 \end{matrix} \qquad \begin{matrix} \parallel \\ \mathbf{z} \end{matrix}$$

$$= \sum_{\alpha=0}^2 \mathbf{z}_\alpha \sigma_\alpha$$

$$|\mathbf{z}_0|^2 + |\mathbf{z}|^2 = \cos^2 \frac{\theta}{2} + \sin^2 \frac{\theta}{2} \underbrace{|\hat{n}|^2}_{=1} = 1$$

Ex: Hadamard transformation.

$$H := \frac{\sigma_x + \sigma_z}{\sqrt{2}} : |H\rangle \longrightarrow |H\rangle.$$

$$\Rightarrow H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \quad \text{in the basis } \{|0\rangle, |1\rangle\}.$$

$$\left\{ \begin{array}{l} H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} \\ H|1\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}} \\ H|x_j\rangle = \frac{|0\rangle + e^{i\pi x_j} |1\rangle}{\sqrt{2}} \quad (x_j \in \{0, 1\}). \end{array} \right.$$

$$H = e^{i\frac{\pi}{2}} D_{\frac{1}{2}}(0) D_{\frac{1}{2}}\left(\frac{\pi}{2}\right) D_{\frac{1}{2}}(-\pi).$$

< Ch3. Tensor products & composite systems >

Definition

let H^A, H^B : Hilbert spaces

$|\varphi\rangle \in H^A$, $|\psi\rangle \in H^B$

Define $|\varphi\rangle \otimes |\psi\rangle : H^A \times H^B \longrightarrow \mathbb{C}$
 $(\xi, \eta) \longmapsto \langle \xi | \varphi \rangle \langle \eta | \psi \rangle$

$\rightarrow$ anti-linear in ξ & η

($\because f := |\varphi\rangle \otimes |\psi\rangle$,

$$f(a\xi + b\xi', \eta) = \langle \underline{a\xi + b\xi'} | \varphi \rangle \langle \eta | \psi \rangle.$$

$$= \overline{a} f(\xi, \eta) + \overline{b} f(\xi', \eta)$$

Similar for η .

$$\boxed{H^A \otimes H^B} := \{ \psi : H^A \times H^B \rightarrow \mathbb{C} \mid \text{anti-linear \& anti } \eta, \exists f \}$$

$\rightarrow \mathbb{C}$ -vector space

$$\left(\begin{aligned} (\psi_1 + \psi_2)(\xi, \eta) &= \psi_1(\xi, \eta) + \psi_2(\xi, \eta) \\ (a\psi)(\xi, \eta) &= a\psi(\xi, \eta) \end{aligned} \right.$$

$$\text{identity: } \circ, \text{ inv } -\psi$$

$$\left. \begin{aligned} &\text{identity: } \circ, \text{ inv } -\psi \end{aligned} \right\}.$$

* Some expressions of $|\psi\rangle \in U(H)$

Properties

let $|\psi\rangle \in H^A$, $|\psi\rangle \in H^B$, $a, b \in \mathbb{C}$

$$1. (a|\psi\rangle) \otimes |\psi\rangle = |\psi\rangle \otimes (a|\psi\rangle) = a(|\psi\rangle \otimes |\psi\rangle)$$

$$\begin{aligned} \therefore (a|\psi\rangle) \otimes |\psi\rangle & \stackrel{(5.7)}{=} \langle \xi | a|\psi\rangle \langle \eta | \psi \rangle \\ & = \underline{a \langle \xi | \psi \rangle \langle \eta | \psi \rangle} \\ & = \langle \xi | \psi \rangle \langle \eta | a|\psi\rangle \end{aligned}$$

$$2. (a+b)|\psi\rangle \otimes |\psi\rangle = a|\psi\rangle \otimes |\psi\rangle + b|\psi\rangle \otimes |\psi\rangle$$

$$3. (|\psi_1\rangle + |\psi_2\rangle) \otimes |\psi\rangle = |\psi_1\rangle \otimes |\psi\rangle + |\psi_2\rangle \otimes |\psi\rangle$$

$$4. |\psi\rangle \otimes (|\psi_1\rangle + |\psi_2\rangle) = |\psi\rangle \otimes |\psi_1\rangle + |\psi\rangle \otimes |\psi_2\rangle.$$

Notation

$$|\psi \otimes \psi\rangle := |\psi\rangle \otimes |\psi\rangle$$

* $H^A \otimes H^B$: Hilbert space

$$1. \text{ For } |\psi_1\rangle, |\psi_2\rangle \in H^A, |\psi_1\rangle, |\psi_2\rangle \in H^B,$$

$$|\psi_i\rangle \otimes |\psi_i\rangle \in H^A \otimes H^B.$$

$$\text{Define } \langle \psi_1 \otimes \psi_1 | \psi_2 \otimes \psi_2 \rangle = \langle \psi_1 | \psi_2 \rangle \langle \psi_1 | \psi_2 \rangle.$$

$$2. \text{ Find basis of } H^A \otimes H^B.$$

let $\{|\psi_i\rangle\}$, $\{|\psi_j\rangle\}$: ONB of H^A & H^B resp.

Consider $\{ |e_a \otimes f_b\rangle \} \subset H^A \otimes H^B$.

$$1) \langle e_a \otimes f_b | e_{a'} \otimes f_{b'} \rangle = \langle e_a | e_{a'} \rangle \langle f_b | f_{b'} \rangle = \delta_{aa'} \delta_{bb'}$$

$\therefore$ orthonormal

$$2) \text{ let } \Psi \in H^A \otimes H^B.$$

$$\begin{aligned} \Psi(\xi, \eta) &= \underline{\Psi} \left(\frac{1}{a} |e_a\rangle \langle e_a| \xi \rangle, \frac{1}{b} |f_b\rangle \langle f_b| \eta \rangle \right) \\ &= \sum_{a,b} \underline{\Psi}(|e_a\rangle, |f_b\rangle) \langle \xi | e_a \rangle \langle \eta | f_b \rangle \\ &= \sum_{a,b} \Psi_{ab} |e_a \otimes f_b\rangle (\xi, \eta) \end{aligned}$$

$$\Psi = \sum_{a,b} \Psi_{ab} |e_a \otimes f_b\rangle$$

$\therefore$ generating set.

$$3) \text{ Suppose } \sum_{a,b} \Psi_{ab} |e_a \otimes f_b\rangle = 0 \text{ where } \Psi_{ab} \in \mathbb{C}$$

$$\Rightarrow \forall (\xi, \eta) \in H^A \times H^B,$$

$$\begin{aligned} \sum_{a,b} \Psi_{ab} (|e_a \otimes f_b\rangle) (\xi, \eta) &= \sum_{a,b} \Psi_{ab} \langle \xi | e_a \rangle \langle \eta | f_b \rangle \\ &= 0. \end{aligned}$$

$$\text{Choose } \xi = |e_{a'}\rangle, \eta = |f_{b'}\rangle$$

$$\begin{aligned} \sum_{a,b} \Psi_{ab} \langle e_{a'} | e_a \rangle \langle f_{b'} | f_b \rangle &= \sum_{a,b} \Psi_{ab} \delta_{aa'} \delta_{bb'} \\ &= \Psi_{a'b'} = 0. \end{aligned}$$

$\therefore$ l. indep.

$\therefore \{ |e_a \otimes f_b\rangle \}$: ONB of $H^A \otimes H^B$.

3. For $|\Psi\rangle, |\Phi\rangle \in H^A \otimes H^B$, where

$$|\Psi\rangle = \sum_{a,b} \Psi_{ab} |e_a \otimes f_b\rangle$$

$$|\Phi\rangle = \sum_{a',b'} \Phi_{a'b'} |e_{a'} \otimes f_{b'}\rangle,$$

$$\begin{aligned} \text{Define } \langle \Psi | \Phi \rangle &:= \sum_{a,b} \sum_{a',b'} \overline{\Psi_{ab}} \Phi_{a'b'} \underbrace{\langle e_a \otimes f_b | e_{a'} \otimes f_{b'} \rangle}_{\substack{\delta_{aa'} \delta_{bb'}}} \\ &= \sum_{a,b} \overline{\Psi_{ab}} \Phi_{ab} \end{aligned}$$